
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Out-of-distribution (OOD) detection plays a crucial role in ensuring the robustness
2 and reliability of machine learning systems deployed in real-world applications.
3 Recent approaches have explored the use of unlabeled data, showing potential for
4 enhancing OOD detection capabilities. However, effectively utilizing unlabeled
5 in-the-wild data remains challenging due to the mixed nature of both in-distribution
6 (InD) and OOD samples. The lack of a distinct set of OOD samples complicates
7 the task of training an optimal OOD classifier. In this work, we introduce Medix, a
8 novel framework designed to identify potential outliers from unlabeled data using
9 the median operation. We use the median because it provides a stable estimate of
10 the central tendency, as an OOD detection mechanism, due to its robustness against
11 noise and outliers. Using these identified outliers, along with labeled InD data,
12 we train a robust OOD classifier. From a theoretical perspective, we derive error
13 bounds that demonstrate Medix achieves a low error rate. Empirical results further
14 substantiate our claims, as Medix outperforms existing methods across the board
15 in open-world settings, confirming the validity of our theoretical insights.

16 1 Introduction

17 Deploying machine learning models in real-world applications often exposes them to challenges
18 related to safety and reliability, particularly due to the presence of out-of-distribution (OOD) data.
19 These OOD samples, stemming from unknown categories, should not be predicted by the model.
20 However, neural networks are inherently vulnerable and typically lack the necessary mechanisms to
21 detect and appropriately handle OOD inputs in practice (Nguyen et al., 2015).

22 Identifying OOD samples during inference is critical yet inherently not easy, as models are not
23 exposed to unknown distributions during training and, therefore, cannot reliably distinguish OOD
24 from in-distribution (InD) data. To address this challenge, recent approaches (Katz-Samuels et al.,
25 2022a; Du et al., 2024a) have explored leveraging additional “in-the-wild” data to improve OOD
26 detection. Specifically, Katz-Samuels et al. (2022a) introduced a method that uses unlabeled wild data
27 for regularizing model training, while still focusing on classifying labeled InD data. The advantage
28 of using such unlabeled wild data lies in its availability—being easily collectible once a model is
29 deployed in its operating environment. This approach allows the model to better capture the true
30 distribution of OOD data encountered during test time, leading to a more robust OOD detection.

31 However, leveraging unlabeled wild data presents significant challenges due to the complex mixture
32 of InD and OOD data. The absence of a distinct and clean set of OOD samples complicates
33 the development of robust OOD detection methods, especially since the OOD detector model
34 only encounters data drawn from this mixed distribution, without knowledge of whether each
35 sample is from the InD or OOD category. At present, the problem remains underexplored, with
36 substantial opportunities for further advancement. Moreover, few studies establish a formal theoretical

foundation, and to the best of our knowledge, Du et al. (2024a) is the only work that provides such a foundation for the “in-the-wild” setting.

Meanwhile, recent studies have demonstrated the effectiveness of median-based approaches in data pruning (Acharya et al., 2024). Prompted by these developments, this paper aims to answer the following question:

How can median-based methods leverage unlabeled wild data to facilitate OOD detection with theoretical guarantees?

In an attempt to provide an affirmative answer to this question, we introduce a novel median-centric perspective for OOD detection. Specifically, we propose a median-based optimization framework and develop an algorithm, Medix, that effectively identifies OOD samples from wild data with low error rate. We provide a theoretical foundation that guarantees minimal error, which is further validated through experiments, demonstrating the robustness and efficiency of our algorithm. We are one of the few studies that provide such a theoretical foundation for the unlabeled “in-the-wild” setting. We show that median-based filtering is robust for outlier detection in unlabeled mixtures by bounding the fraction of InD samples incorrectly flagged as outliers. This fraction is controlled by two effects: the contamination effect, which quantifies the impact of OOD points and remains manageable as long as OOD proportion is below 50%; and the concentration effect, which leverages the sub-Gaussian nature of InD gradients to bound deviations of InD points from their mean gradient. Together, these effects ensure that, with high probability, the number of misclassified InD points remains small, demonstrating the method’s effectiveness even in the worst-case scenario.

We benchmark our approach against two categories of methods: (1) those trained solely on InD data, and (2) those trained with both InD data and an auxiliary unlabeled dataset. On CIFAR-100 (Krizhevsky et al., 2009), Medix demonstrates a significant improvement over the strong baseline KNN+ (Sun et al., 2022), outperforming it by an average of 40.98% in terms of FPR95. Unlike approaches such as Outlier Exposure (Hendrycks et al., 2019), which rely on a clean, auxiliary unlabeled dataset (i.e. they make a strong distributional assumption that the auxiliary data is completely separable from the InD data), Medix achieves superior results without such assumptions, offering greater flexibility. Compared to WOODS (Katz-Samuels et al., 2022a), Medix reduces the average FPR95 by 1.32% on CIFAR-100 and 2.60% on CIFAR-10.

Our key contributions are as follows:

- C1)** We propose Medix, a median-centric greedy approach that filters outliers from the unlabeled wild data and then trains an OOD detector on the identified outliers and the InD samples. Our main contribution is the filtering stage.
- C2)** We establish theoretical guarantees for the robustness of median-based filtering in identifying both inliers and outliers within unlabeled mixtures. Specifically, we prove that the misclassification rates for both InD samples flagged as outliers and OOD samples retained as inliers are tightly controlled by two effects: the contamination effect, which remains bounded as long as the proportion of OOD samples is below 50%, and the concentration effect, which ensures stability in the gradient behavior of InD samples.
- C3)** We conduct an extensive evaluation of Medix across eleven InD-OOD pairs, comparing its performance against 20 competitive baselines. Our results demonstrate that Medix outperforms all the baselines, achieving superior performance across the board. We show via experiments that Medix outlier extraction achieves a low error rate (e.g. only 12.5%; see Figure 2), corroborating our theoretical findings.

2 Preliminaries and Problem Setup

In this section, we first provide an overview of the OOD detection problem, and then formally define the data setup, model architecture, loss functions, and the learning goal.

Labeled In-Distribution Data. Consider the input space $\mathcal{X}$ and the label space $\mathcal{Y} = \{1, \dots, K\}$, which together define the structure of InD data. A labeled dataset $\mathcal{S}_{\text{in}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ is generated by sampling n pairs independently and identically distributed (i.i.d.) from $\mathbb{P}_{XY}$, an unknown joint probability distribution over $\mathcal{X} \times \mathcal{Y}$. The marginal distribution of $\mathbb{P}_{XY}$ on $\mathcal{X}$ is denoted as $\mathbb{P}_{\text{in}}$, representing the underlying distribution of InD inputs. We use $\mathcal{S}_{\text{in}}$ to train an InD model.

87 **Out-of-distribution detection.** We address a practical scenario where the model is trained using
 88 labeled InD data but is later deployed in environments that may contain OOD inputs from classes not
 89 represented in the training data, i.e., for some label $y \notin \mathcal{Y}$. The model is expected to abstain from
 90 making predictions for such OOD inputs. At inference time, the primary objective is to determine
 91 whether a given input belongs to the InD distribution or arises from an OOD source.

92 **Unlabeled wild data.** One of the primary obstacles in OOD detection is the scarcity of labeled
 93 OOD samples. The potential sample space of OOD data can be very large, making the collection
 94 of labeled examples both costly and impractical. To address this, we introduce unlabeled wild data,
 95 $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, into our learning framework to better mimic real-world scenarios as proposed
 96 by Katz-Samuels et al. (2022a). Wild data is a blend of InD and OOD samples and can be readily
 97 collected during the deployment phase of a pre-trained model on $\mathcal{S}_{\text{in}}$. Similar to Du et al. (2024a);
 98 Katz-Samuels et al. (2022a), we adopt the Huber contamination model (Huber, 1964) to characterize
 99 the marginal distribution of the wild data

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

100 where π denotes the contamination proportion and $\mathbb{P}_{\text{out}}$ captures the OOD distribution over $\mathcal{X}$. We
 101 note that the scenario where $\pi = 0$ corresponds to the absence of OOD samples, rendering the
 102 problem trivial.

103 **Models and Loss Functions.** Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ represent the InD classifier parameterized by
 104 $\phi \in \Phi$, where Φ denotes the parameter space for this classifier. The output of f_ϕ corresponds to a
 105 soft probability distribution over the $K = |\mathcal{Y}|$ InD classes. The loss function for the labeled InD data
 106 is defined as $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For OOD detection, we introduce a separate classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$,
 107 parameterized by $\theta \in \Theta$, with Θ as the parameter space. The binary loss function associated with
 108 g_θ is denoted as $\ell_b(g_\theta(x), y_b)$, where $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$. Here, $y_+ > 0$ represents the InD class,
 109 while $y_- < 0$ corresponds to the OOD class.

110 **Learning objective.** Our learning framework is designed to simultaneously train the OOD detector
 111 g_θ and the multi-class classifier f_ϕ , leveraging both the InD data $\mathbb{P}_{\text{in}}$ and the wild data $\mathbb{P}_{\text{wild}}$. During
 112 testing, we evaluate the performance using the following metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{\mathcal{X}\mathcal{Y}}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

113 where $\mathbb{I}\{\cdot\}$ denotes the indicator function, and $\mathbb{P}_{\text{out}}^{\text{test}}$ represents the OOD test data distribution.

114 3 Method: Median-Centric Framework for OOD Detection

115 In this section, we present a novel learning paradigm, termed **Medix**, designed for OOD detection by
 116 harnessing the power of unlabeled wild data. Our framework overcomes the limitations of conven-
 117 tional approaches that rely exclusively on InD data and is particularly well-suited for applications in
 118 open-world environments, where models are often confronted with previously unseen inputs. The
 119 Medix framework is composed of two integral stages: **1) Outlier Extraction:** A filtering process
 120 that isolates candidate OOD samples from the unlabeled wild data (explained in Section 3.1), and **2)**
 121 **Detector Training:** train a binary OOD detector using both InD data and the outlier candidates identi-
 122 fied in the previous step (explained in Section 3.2). For stage 2, we follow the protocol introduced by
 123 Du et al. (2024a). As we will demonstrate in the subsequent sections, this two-step methodology not
 124 only facilitates the effective extraction of OOD data from the unlabelled wild data but also establishes
 125 a robust foundation for deploying machine learning models in dynamic, open-world scenarios.

126 3.1 Extracting Candidate Outliers from the Wild Data

127 To isolate potential outliers from the wild mixture $\mathcal{S}_{\text{wild}}$, our framework leverages an optimization-
 128 based approach that exploits the gradients of the model parameters. These gradients are derived from
 129 a classification model, f_ϕ , which is trained solely on the InD dataset $\mathcal{S}_{\text{in}}$. The detailed methodology
 130 for this process is formally outlined below.

131 **Reference gradient estimation from InD data.** The first step in our proposed framework is to
 132 estimate a reference gradient using the InD dataset $\mathcal{S}_{\text{in}}$. This is achieved by training a classifier f_ϕ on

133 $\mathcal{S}_{\text{in}}$ through empirical risk minimization (ERM) as follows

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_{\phi}), \text{ where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_{\phi}) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_{\phi}(\mathbf{x}_i), y_i), \quad (2)$$

134 where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. Once the classifier has been trained, we compute the mean
135 gradient $\bar{\nabla}_{\text{in}}$ as the average of the gradients of the loss function with respect to the model parameters
136 over the InD data:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\mathbf{x}_i), y_i). \quad (3)$$

137 In our approach, $\bar{\nabla}_{\text{in}}$ serves as the reference gradient, enabling the quantification of deviations for
138 other data points relative to this reference.

139 **Motivation.** We hypothesize that increasing the number of OOD samples in the wild dataset, $\mathcal{S}_{\text{wild}}$,
140 will lead to a greater deviation from the average InD gradient, $\bar{\nabla}_{\text{in}}$. To test this hypothesis, we
141 design an initial experiment using CIFAR-10 (Krizhevsky et al., 2009) as the InD dataset and SVHN
142 (Netzer et al., 2011) as the OOD dataset. Specifically, $\mathcal{S}_{\text{wild}}$ consists of 10,000 samples drawn from
143 CIFAR-10, ensuring that these samples are disjoint from the training set used to train the model $\phi_{\mathcal{S}_{\text{in}}}$,
144 which we leverage to compute $\bar{\nabla}_{\text{in}}$. We incrementally add SVHN OOD samples to $\mathcal{S}_{\text{wild}}$ and track
145 the behavior of the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of the
146 gradients of the wild dataset as follows:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

147 Our results, depicted in Figure 1, reveal a clear and monotonic increase in the L_2 -norm deviation,
148 supporting our hypothesis. This observation serves as a key motivation for the method we introduce
149 in the subsequent section. Notably, the stopping criterion for our algorithm is derived from this
150 monotonically increasing behavior, where we terminate the algorithm when the L_2 -norm deviation
151 between consecutive iterations drops below a threshold ϵ ; we will explain this method in detail in the
152 following section.

153 **Filtering potential outliers from unlabeled wild data.** Motivated by the results in Figure 1, we
154 formulate the following optimization problem to identify the outlier subset $\mathcal{S}_{\text{out}}^*$ in $\mathcal{S}_{\text{wild}}$:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

155 Here $\text{EWM}(\cdot)$ denotes the element-wise median function, and $\hat{y}_{\tilde{\mathbf{x}}_i}$ represents the predicted
156 label for a wild sample $\tilde{\mathbf{x}}_i$. For notational simplicity, we define $G_{\mathcal{S}} = \{\nabla \ell(\tilde{\mathbf{x}}_i)\}_{i \in \mathcal{S}}$.
157 The above optimization problem aims to identify
158 a subset $\mathcal{S}$ in $\mathcal{S}_{\text{wild}}$ that minimizes the distance be-
159 tween the EWM of the gradients and the average
160 gradient $\bar{\nabla}_{\text{in}}$. According to Figure 1, such a subset,
161 denoted by $\mathcal{S}_{\text{in}}^*$, may well represent the InD data in
162 $\mathcal{S}_{\text{wild}}$, in which case $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ may capture
163 the OOD data in $\mathcal{S}_{\text{wild}}$.

164 Solving the optimization problem in (4) can be
165 computationally prohibitive, especially as the size
166 of $\mathcal{S}_{\text{wild}}$ increases. To address this, we propose a
167 greedy approximation based on a leave-one-out ap-
168 proach, as outlined in Algorithm 1. The algorithm
169 implements an iterative procedure for outlier detec-
170 tion from a wild dataset, leveraging deviations in gradient information relative to the InD dataset.

171 The algorithm begins by computing the EWM of the wild data gradients at each iteration, which
172 serves as a reference for comparing against the average gradient of the InD data $\bar{\nabla}_{\text{in}}$. We denote
173 by d_t , the L_2 distance between the average InD gradient $\bar{\nabla}_{\text{in}}$ and the EWM gradients of the data
174 left in the wild set, represented by $\mathcal{S}$, i.e., $d_t = \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$. The algorithm then iteratively
175 identifies samples in $\mathcal{S}$ that incur the most significant drop in the L_2 distance with d_t when removed

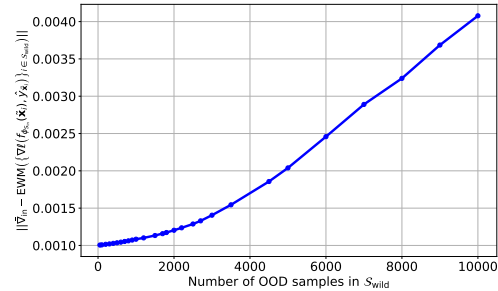


Figure 1: Distance deviation as we increase OOD samples in $\mathcal{S}_{\text{wild}}$.

from $\mathcal{S}$ as OOD. Specifically, having data samples $\mathcal{S}$ left, we remove each sample $i \in \mathcal{S}$ and compute the EWM as $\text{EWM}(G_{\mathcal{S} \setminus \{i\}})$ and the distance to $\bar{\nabla}_{\text{in}}$ as $\|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$. We then find the drop in distance $\delta_i = d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$ and find the k samples with the largest drop and identify them as OOD. The algorithm repeats until there is no significant drop in δ_i or a maximum number of iterations is reached. The convergence criterion is based on the change in the L_2 distance between two iterations, which must fall below a predefined ϵ threshold; this criterion is inspired by the monotonically increasing trend that we observed in our preliminary experiment in Figure 1, i.e., as a substantial number of OOD samples are removed, the L_2 distance gradually decreases to a small value, signaling the point at which the algorithm should halt to avoid erroneously identifying InD samples as outliers, thus preventing any degradation in performance.

Algorithm 1 Iterative Outlier Detection via **Medix**

Require: $\bar{\nabla}_{\text{in}}$, $G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i})$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$ (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\hat{\mathcal{S}}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

3.2 Training the OOD Detector with Candidate Outliers

After identifying the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$ from the wild data, we proceed to train the OOD detector g_θ designed to maximize the distinction between InD and candidate outlier data following the protocol in Du et al. (2024a). The objective function explicitly enforces separability at the decision boundary (thresholded at 0), assigning positive outputs for labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ and negative outputs for candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$. Specifically, the loss function is defined as:

$$\begin{aligned} \mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_\theta) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_\theta(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_\theta(\tilde{\mathbf{x}}) > 0\}. \end{aligned} \tag{5}$$

To address the challenge posed by the non-differentiable nature of the 0/1 loss, a binary loss based on a differentiable sigmoid function is employed as a smooth surrogate, ensuring tractable optimization while retaining alignment with the original objective. The learning framework incorporates the InD risk as defined in Equation 2, thereby safeguarding the predictive accuracy for InD samples. This unified optimization approach significantly bolsters the generalization performance of g_θ , enabling robust detection of OOD samples drawn from $\mathbb{P}_{\text{out}}$, as corroborated by our results.

4 Theoretical Analysis

We now present the theoretical guarantees of Medix’s filtering stage. The following theorems provide provable upper bounds on the misclassification rates for both InD and OOD points. Together, they demonstrate the two-sided robustness of our EWM filtering. For detailed proofs, see Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points*

in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

Theorem 4.1 reveals that the inlier misclassification rate is governed by two key effects: a *concentration term*, which vanishes with larger sample size and captures statistical fluctuations in the InD gradients, and a *contamination term*, which quantifies the influence of OOD points on the EWM. Notably, the bound remains controlled as long as the contamination ratio $\pi < 0.5$, underscoring the robustness of median-based filtering to moderate OOD presence. We now turn to the complementary question: *how many true outliers fail to be flagged and are mistakenly retained?* The next theorem answers this by bounding the OOD misclassification rate.

Theorem 4.2 (Outlier Misclassification Bound). Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1-\pi}{2\pi}}_{\text{Contamination term}}.$$

Theorem 4.2 complements the previous result by bounding the fraction of OOD points mistakenly retained as InD. Together, Theorems 4.1 and 4.2 provide a two-sided guarantee for the robustness of the Medix filtering method. The inlier and outlier misclassification rates are governed by a balance between the following three effects:

Contamination Effect. For Theorem 4.1, even when the wild dataset contains a significant fraction π of OOD points, the median-based criterion remains stable as long as $\pi < 0.5$. This is reflected in the $\pi/[2(1-\pi)]$ term of the bound, which increases with π and encodes the risk that OOD gradients may skew the EWM. Conversely, Theorem 4.2 includes a symmetric penalty $\frac{1-\pi}{2\pi}$, quantifying the difficulty of isolating OOD points when they are underrepresented.

Concentration Effect. Both bounds exploit the sub-Gaussian nature of the gradient coordinates. For InD data, this ensures that most gradients concentrate near $\bar{\nabla}_{\text{in}}$, keeping the median stable even in finite samples. For OOD points, concentration around μ_{out} allows us to quantify the risk that a misaligned OOD sample slips past the filter. These concentration effects decay as $1/\sqrt{m}$, where m is the number of samples from each class.

Separation Effect. Unique to Theorem 4.2, this effect quantifies how far the OOD mean gradient must lie from the InD mean gradient in order to reliably reject OOD samples. The exponential term $\exp(-(\Delta - \epsilon)^2/2\sigma_{\text{out}}^2)$ captures this trade-off: the more separated the distributions, the less likely it is for OOD gradients to fool the filter.

Theorems 4.1 and 4.2 jointly establish that, with high probability, median filtering achieves robust separation of InD and OOD samples in unlabeled data mixtures. The fraction of InD samples misclassified as outliers is bounded by contamination and concentration effects, while the fraction of OOD points incorrectly retained is governed by an exponential separation term, a concentration bound, and a reverse contamination effect. These results provide rigorous theoretical assurance that Medix minimizes both types of errors under mild assumptions.

Remark 4.3. *Sub-Gaussian Assumption for Gradient Coordinates.* The assumption that gradients of InD samples in each coordinate are sub-Gaussian is crucial for deriving concentration bounds

and ensuring robustness in statistical estimation. Further empirical evidence as shown in Figure 4a confirms this assumption, where we observe that the histogram of gradient values for InD samples is bell-shaped and concentrated around the mean. This indicates that the gradients exhibit light tails, a defining characteristic of sub-Gaussian random variables, in which variables have exponentially decaying tails, meaning extreme gradient values are rare. This aligns with the observed behavior in the histogram. Furthermore, Figure 4b compares the empirical quantiles of the gradient distribution with the theoretical quantiles of a Gaussian distribution in a Q-Q plot (Wilk and Gnanadesikan, 1968). The close alignment of points with the 45-degree reference line demonstrates that the empirical distribution of gradients indeed closely resembles a Gaussian distribution. Since Gaussian random variables are a subset of sub-Gaussian variables, this supports the sub-Gaussian assumption. *We also provide a looser version of these bounds, which removes the sub-Gaussian assumption and holds under merely bounded second moments. This version is presented in Theorem C.3 (Appendix C.3). While the rates degrade, the core robustness guarantee of Medix still holds in this broader setting.*

5 Experiments

This section will demonstrate the efficacy of Medix across various InD-OOD dataset pairs, benchmarking it against 20 widely-used baselines. All experiments are performed on hardware equipped with NVIDIA A100-SXM4-80GB GPUs. The necessary code to reproduce our results will be released upon acceptance.

5.1 Models, Datasets, and Baselines

Datasets. We use the same experimental protocol as Katz-Samuels et al. (2022a), which introduced the problem of learning OOD detectors with wild data, enabling a fair comparison to prior work. Specifically, we use CIFAR-10 and CIFAR-100 as InD datasets ($\mathbb{P}_{\text{in}}$). For OOD testing, we select a suite of natural image datasets, including PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi et al., 2014), and LSUN-RESIZE & LSUN-C (Yu et al., 2015) as OOD datasets ($\mathbb{P}_{\text{out}}$). To simulate wild data ($\mathbb{P}_{\text{wild}}$), we combine a subset of InD data ($\mathbb{P}_{\text{in}}$) with the OOD data ($\mathbb{P}_{\text{out}}$) under a default mixing parameter $\pi = 0.5$. For example, when using PLACES365 as an OOD test set, we construct a wild mixture by combining CIFAR with PLACES365 as wild data and test on PLACES365 as the OOD set. This procedure is repeated across all OOD datasets and baselines. The InD CIFAR dataset is split into two halves: the first 25,000 images to train $\phi_{\mathcal{S}_{\text{in}}}$, while the remaining images to generate the wild mixture $\mathcal{S}_{\text{wild}}$. For gradient computation, we use the penultimate layer weights, as these have been shown to be particularly informative for OOD detection (Huang et al., 2021a).

Evaluation metrics. We evaluate using three standard metrics: (1) False Positive Rate (FPR95↓) of OOD samples when the True Positive Rate of InD samples is 95%, (2) Area Under the Receiver Operating Characteristic Curve (AUROC↑), and (3) InD Classification Accuracy (InD Acc↑).

Baselines. Our comparison encompasses a diverse set of competitive baselines, categorized based on whether they are trained using only InD data or both InD and wild data. For the methods trained exclusively on InD data ($\mathbb{P}_{\text{in}}$), we compare Medix against a variety of established OOD detection methods, including Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022); these methods use a model trained with softmax cross-entropy loss. Additionally, we also compare against methods based on contrastive loss, such as CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022), for a more comprehensive comparison. For methods that leverage both InD and wild data, we compare against Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularization learning (Liu et al., 2020b), which regularize the training by promoting lower confidence or higher energy on outlier data. We also include a comparison with WOODS (Katz-Samuels et al., 2022a), which introduced the concept of wild unlabeled data and utilizes it for OOD detection through a constrained optimization approach. Finally, we included more recent baselines, including CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024), to provide a more thorough evaluation.

5.2 Experimental Setup

In line with WOODS (Katz-Samuels et al., 2022a), we employ a Wide ResNet architecture (Zagoruyko, 2016) with 40 layers and a width factor of 2 for the InD classifier $\phi_{S_{in}}$. It is trained using stochastic gradient descent with a momentum of 0.9, weight decay of 0.0005, and an initial learning rate of 0.1. Training is performed for 100 epochs with cosine learning rate decay, a batch size of 128, and a dropout rate of 0.3. Hyperparameters ϵ and k used in the proposed method Medix are selected from the sets $\{5e-5, 5e-4, 5e-3, 5e-2\}$ and $\{4k, 7k, 10k, 20k\}$, respectively, taking into account dataset sizes and with the objective of maximizing OOD performance. For the OOD classifier g_θ , we initialize it with the pre-trained InD classifier $\phi_{S_{in}}$ and add a linear layer that performs binary classification using the penultimate-layer features. The learning rate for this classifier is set to 0.001, and fine-tuning is done for 100 epochs as outlined in Equation (5). We combine the binary classification loss with the InD classification loss, assigning a weight of 10 to the binary classification component. All other training parameters remain the same as those used for training $\phi_{S_{in}}$.

Table 1: OOD detection performance comparison of Medix and baselines on CIFAR-10 as InD data. Performance averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_{in}$ only													
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	40.76	89.57	41.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
Using $\mathbb{P}_{in}$ and $\mathbb{P}_{wild}$													
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

Table 2: OOD detection performance comparison of Medix and baselines on CIFAR-100 as InD data. Performance averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets												InD Acc $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_{in}$ only													
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78
Using $\mathbb{P}_{in}$ and $\mathbb{P}_{wild}$													
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83

5.3 Results

We present our main results in Table 2 on CIFAR-100, where Medix remarkably outperforms all OOD detection baselines. The results highlight the following key observations: (1) Methods trained on both InD and wild data significantly outperform those trained exclusively on InD data. Medix reduces the FPR95 by 52.31% on PLACES365 and 38.24% on TEXTURES compared to KNN+, demonstrating the effectiveness of incorporating in-the-wild data for model regularization. (2) Medix further outperforms competitive methods utilizing wild data ($\mathbb{P}_{wild}$): On CIFAR-100, Medix achieves an average FPR95 of 5.42%, which represents a 1.32% improvement over WOODS. Additionally, Medix maintains a competitive InD accuracy of 73.33%. This slight difference can be attributed to the fact that our method is trained on 25,000 labeled InD samples, while baseline methods, which do not leverage wild data, use the full CIFAR-100 training set of 50,000 samples. We present the results for CIFAR-10 in Table 1, where Medix surpasses all baseline methods. Medix outperforms

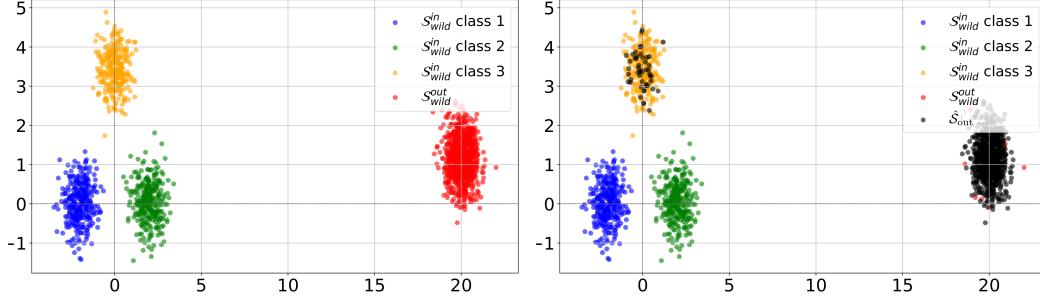


Figure 2: Example of Medix applied to unlabeled wild data. (a) Setup of the InD data $\mathcal{S}_{\text{wild}}^{\text{in}}$ and OOD data $\mathcal{S}_{\text{wild}}^{\text{out}}$ in the wild, with inliers sampled from three multivariate Gaussian distributions. (b) Outliers $\hat{\mathcal{S}}_{\text{out}}$ filtered by Medix (in black), with an error rate of $\hat{\mathcal{S}}_{\text{out}}$ containing InD data $\mathcal{S}_{\text{wild}}^{\text{in}}$ is only 12.5%.

WOODS by 7.21% on PLACES365 and 5.25% on TEXTURES in terms of FPR95, demonstrating its effectiveness in detecting OOD samples.

A representative visual example of Medix. We further investigate the performance of Algorithm 1 (Outlier Extraction) in extracting OOD samples from wild data $\mathcal{S}_{\text{wild}}$. To visualize this, we design an experiment using 2-dimensional synthetic data. This simulation is designed to be simple to facilitate better understanding. We generate the InD data by sampling from three multivariate Gaussian distributions, corresponding to three classes. The mean vectors are set to $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$, respectively. The covariance matrix for all three classes is fixed at $0.25 \cdot I$. For the OOD data, we use a multivariate Gaussian distribution $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. In Figure 2, we observe that our method successfully identifies outliers, the majority of which align with the ground truth. Medix successfully flags 87.5% of actual OOD samples as outliers, underscoring its robustness in outlier extraction.

Additional studies and insights. Due to space constraints, we defer additional experiments and insights to Appendix A, which include (1) ablation studies on hyperparameter selection and sensitivity analysis (Appendix A.2), showing Medix’s strong robustness to hyperparameters, (2) a comparison between EWM and geometric median (Appendix A.1), showing that EWM is more sensitive to distributional shifts, making it more effective and reliable choice for filtering in our method, (3) a comprehensive comparison with methods employing competitive contrasting learning objectives (Appendix A.3), demonstrating that Medix outperforms even these competitive baselines by a significant margin, (4) evaluation on large-scale data under the complex unseen OOD setting, where $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$ (Appendix A.4), demonstrating that Medix outperforms baselines by a significant margin, (5) evaluating computation and memory efficiency of Medix (Appendix A.6) (6) evaluating the impact of pseudo-label quality (Appendix A.5), showing that our method is resilient to noisy or low-confidence labels, and (4) a comparison with semi-supervised open-set recognition methods (Appendix A.7).

6 Related Work

In recent years, there has been a growing interest in OOD detection (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djurisić et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). One approach to detect OOD data uses scoring functions to assess data distribution, including distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Schwag et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based score (Huang et al., 2021b), energy-based score (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based approaches (Bendale and Boult, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020).

Another approach to OOD detection involves using regularization techniques during the training phase (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein,

2020; Jeong and Kim, 2020; Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a). For example, regularization techniques can be applied to the model to either reduce its confidence (Lee et al., 2017; Hendrycks et al., 2019) or increase its energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on the OOD data. Most of these regularization methods assume the availability of an auxiliary OOD dataset.

Several studies (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023) have relaxed the assumption of using only labeled data by incorporating unlabeled wild data (Katz-Samuels et al., 2022a; Geng et al., 2025), though they did not propose a clear mechanism for outlier detection. In contrast, Du et al. (2024a,b) introduced an explicit outlier filtering method, but their thresholding technique differs fundamentally from ours, as we utilize a new median-centric approach to detect the outliers. Additionally, Katz-Samuels et al. (2022a); Du et al. (2024a) operate under the assumption of batch-level mixing, where each batch has a set ratio of InD and OOD samples. However, with large outsourced datasets, such structured mixing is not available—data is mixed randomly across the dataset. Our method addresses this by enabling dataset-level mixing without relying on batch-level structure. Many studies also leverage positive-unlabeled learning, which trains classifiers using positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022). However, a key distinction from our approach is that these methods focus solely on differentiating $\mathbb{P}_{\text{out}}$ from $\mathbb{P}_{\text{in}}$, without simultaneously training an OOD classifier. Additionally, we propose a median-centric method to identify outliers in unlabeled data, with provably low error rates.

7 Conclusions

In this work, we introduced Medix, a novel median-centric framework for OOD detection that leverages unlabeled in-the-wild data. Using the inherent robustness of median operation, Medix effectively filters outliers from mixed unlabeled data, enabling the training of a reliable OOD detector. Our theoretical analysis established provable bounds on the inlier misclassification rate, demonstrating that Medix maintains robustness even under significant OOD contamination (up to 50%), with errors controlled by sub-Gaussian concentration and contamination effects. We also provided complementary theoretical limits on the rate of OOD misclassification to accurately isolate OOD samples under clear separation conditions. Empirical validation across diverse benchmarks showcased Medix’s performance superiority over 20 baselines: it reduced the average FPR95 by 40.98% compared to strong baselines like KNN+ and outperformed state-of-the-art methods such as WOODS and DRL, while achieving an outlier extraction error rate as low as 12.5%.

References

- Abbas, M., Azmat, M., Horesh, R., and Yurochkin, M. (2025). Out-of-distribution detection using synthetic data generation. In *Second Conference on Language Modeling*.
- Acharya, A., Dhillon, I. S., and Sanghavi, S. (2024). Geometric median (gm) matching for robust data pruning. *arXiv preprint arXiv:2406.17188*.
- Acharya, A., Sanghavi, S., Jing, L., Bhushanam, B., Choudhary, D., Rabbat, M., and Dhillon, I. (2022). Positive unlabeled contrastive learning. *arXiv preprint arXiv:2206.01206*.
- Bai, H., Canal, G., Du, X., Kwon, J., Nowak, R. D., and Li, Y. (2023). Feed two birds with one scone: Exploiting wild data for both out-of-distribution generalization and detection. In *International Conference on Machine Learning*, pages 1454–1471. PMLR.
- Bendale, A. and Boulton, T. E. (2016). Towards open set deep networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1563–1572.
- Chapel, L., Alaya, M. Z., and Gasso, G. (2020). Partial optimal transport with applications on positive-unlabeled learning. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing Systems*, volume 33, pages 2903–2913. Curran Associates, Inc.
- Cimpoi, M., Maji, S., Kokkinos, I., Mohamed, S., and Vedaldi, A. (2014). Describing textures in the wild. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3606–3613.

394 Djuricic, A., Bozanic, N., Ashok, A., and Liu, R. (2022). Extremely simple activation shaping for out-of-
395 distribution detection. *arXiv preprint arXiv:2209.09858*.

396 Djuricic, A., Bozanic, N., Ashok, A., and Liu, R. (2023). Extremely simple activation shaping for out-of-
397 distribution detection. In *The Eleventh International Conference on Learning Representations*.

398 Du, X., Fang, Z., Diakonikolas, I., and Li, Y. (2024a). How does unlabeled data provably help out-of-distribution
399 detection? In *The Twelfth International Conference on Learning Representations*.

400 Du, X., Ghosh, R., Sim, R., Salem, A., Carvalho, V., Lawton, E., Li, Y., and Stokes, J. W. (2024b). Vlmguard:
401 Defending vlms against malicious prompts via unlabeled data. *arXiv preprint arXiv:2410.00296*.

402 Du, X., Gozum, G., Ming, Y., and Li, Y. (2022a). Siren: Shaping representations for detecting out-of-distribution
403 objects. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in*
404 *Neural Information Processing Systems*, volume 35, pages 20434–20449. Curran Associates, Inc.

405 Du, X., Sun, Y., Zhu, J., and Li, Y. (2023). Dream the impossible: Outlier imagination with diffusion models.
406 In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine, S., editors, *Advances in Neural*
407 *Information Processing Systems*, volume 36, pages 60878–60901. Curran Associates, Inc.

408 Du, X., Wang, X., Gozum, G., and Li, Y. (2022b). Unknown-aware object detection: Learning what you don’t
409 know from videos in the wild. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern*
410 *Recognition*, pages 13678–13688.

411 Du, X., Wang, Z., Cai, M., and Li, S. (2022c). Towards unknown-aware learning with virtual outlier synthesis.
412 In *International Conference on Learning Representations*.

413 Fan, Y., Kukleva, A., Dai, D., and Schiele, B. (2023). Ssb: Simple but strong baseline for boosting performance
414 of open-set semi-supervised learning. In *Proceedings of the IEEE/CVF International Conference on Computer*
415 *Vision (ICCV)*, pages 16068–16078.

416 Fang, Z., Li, Y., Lu, J., Dong, J., Han, B., and Liu, F. (2022). Is out-of-distribution detection learnable? In
417 Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural*
418 *Information Processing Systems*, volume 35, pages 37199–37213. Curran Associates, Inc.

419 Fort, S., Ren, J., and Lakshminarayanan, B. (2021). Exploring the limits of out-of-distribution detection.
420 *Advances in Neural Information Processing Systems*, 34:7068–7081.

421 Gal, Y. and Ghahramani, Z. (2016). Dropout as a bayesian approximation: Representing model uncertainty
422 in deep learning. In Balcan, M. F. and Weinberger, K. Q., editors, *Proceedings of The 33rd International*
423 *Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 1050–
424 1059, New York, New York, USA. PMLR.

425 Galil, I., Dabbah, M., and El-Yaniv, R. (2023). A framework for benchmarking class-out-of-distribution detection
426 and its application to imagenet. In *The Eleventh International Conference on Learning Representations*.

427 Garg, S., Balakrishnan, S., and Lipton, Z. (2022). Domain adaptation under open set label shift. In Koyejo,
428 S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural Information*
429 *Processing Systems*, volume 35, pages 22531–22546. Curran Associates, Inc.

430 Garg, S., Wu, Y., Smola, A. J., Balakrishnan, S., and Lipton, Z. (2021). Mixture proportion estimation and
431 pu learning: a modern approach. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan,
432 J. W., editors, *Advances in Neural Information Processing Systems*, volume 34, pages 8532–8544. Curran
433 Associates, Inc.

434 Geifman, Y. and El-Yaniv, R. (2019). SelectiveNet: A deep neural network with an integrated reject option. In
435 Chaudhuri, K. and Salakhutdinov, R., editors, *Proceedings of the 36th International Conference on Machine*
436 *Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2151–2159. PMLR.

437 Geng, C., Li, Q., Wang, X., Liang, D., Chen, S., and Yuen, P. C. (2025). Lod: Loss-difference ood detection by
438 intentionally label-noisifying unlabeled wild data. *arXiv preprint arXiv:2505.12952*.

439 Ghosal, S. S., Sun, Y., and Li, Y. (2024). How to overcome curse-of-dimensionality for out-of-distribution
440 detection? In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19849–19857.

441 Gong, T., Wang, G., Ye, J., Xu, Z., and Lin, M. (2018). Margin based pu learning. In *Proceedings of the AAAI*
442 *Conference on Artificial Intelligence*, volume 32.

443 Hang, J.-Y. and Zhang, M.-L. (2024). Binary decomposition: A problem transformation perspective for open-set
444 semi-supervised learning. In Salakhutdinov, R., Kolter, Z., Heller, K., Weller, A., Oliver, N., Scarlett, J., and
445 Berkenkamp, F., editors, *Proceedings of the 41st International Conference on Machine Learning*, volume 235
446 of *Proceedings of Machine Learning Research*, pages 17505–17518. PMLR.

447 He, R., Li, R., Han, Z., Yang, X., and Yin, Y. (2023). Topological structure learning for weakly-supervised
448 out-of-distribution detection. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages
449 4858–4866.

450 Hein, M., Andriushchenko, M., and Bitterwolf, J. (2019). Why relu networks yield high-confidence predictions
451 far away from the training data and how to mitigate the problem. In *Proceedings of the IEEE/CVF conference*
452 *on computer vision and pattern recognition*, pages 41–50.

453 Hendrycks, D. and Gimpel, K. (2016). A baseline for detecting misclassified and out-of-distribution examples in
454 neural networks. *arXiv preprint arXiv:1610.02136*.

455 Hendrycks, D. and Gimpel, K. (2017). A baseline for detecting misclassified and out-of-distribution examples in
456 neural networks. In *International Conference on Learning Representations*.

457 Hendrycks, D., Mazeika, M., and Dietterich, T. (2019). Deep anomaly detection with outlier exposure. In
458 *International Conference on Learning Representations*.

459 Hsieh, C.-J., Natarajan, N., and Dhillon, I. (2015). Pu learning for matrix completion. In Bach, F. and Blei, D.,
460 editors, *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings*
461 *of Machine Learning Research*, pages 2445–2453, Lille, France. PMLR.

462 Huang, R., Geng, A., and Li, Y. (2021a). On the importance of gradients for detecting distributional shifts in the
463 wild. *Advances in Neural Information Processing Systems*, 34:677–689.

464 Huang, R., Geng, A., and Li, Y. (2021b). On the importance of gradients for detecting distributional shifts in the
465 wild. In Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W., editors, *Advances in Neural Information*
466 *Processing Systems*.

467 Huber, P. J. (1964). Robust estimation of a location parameter. *The Annals of Mathematical Statistics*, 35(1):73–
468 101.

469 Jeong, T. and Kim, H. (2020). Ood-maml: Meta-learning for few-shot out-of-distribution detection and
470 classification. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in*
471 *Neural Information Processing Systems*, volume 33, pages 3907–3916. Curran Associates, Inc.

472 Katz-Samuels, J., Nakhleh, J. B., Nowak, R., and Li, Y. (2022a). Training ood detectors in their natural habitats.
473 In *International Conference on Machine Learning*, pages 10848–10865. PMLR.

474 Katz-Samuels, J., Nakhleh, J. B., Nowak, R., and Li, Y. (2022b). Training OOD detectors in their natural
475 habitats. In Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S., editors, *Proceedings*
476 *of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning*
477 *Research*, pages 10848–10865. PMLR.

478 Kim, S., Kim, D., Cho, M., and Kwak, S. (2020). Proxy anchor loss for deep metric learning. In *2020 IEEE/CVF*
479 *Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3235–3244.

480 Kristiadi, A., Hein, M., and Hennig, P. (2020). Being bayesian, even just a bit, fixes overconfidence in relu
481 networks. In *International conference on machine learning*, pages 5436–5446. PMLR.

482 Krizhevsky, A., Hinton, G., et al. (2009). Learning multiple layers of features from tiny images.

483 Lakshminarayanan, B., Pritzel, A., and Blundell, C. (2017). Simple and scalable predictive uncertainty estimation
484 using deep ensembles. In Guyon, I., Luxburg, U. V., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S.,
485 and Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates,
486 Inc.

487 Lee, K., Lee, H., Lee, K., and Shin, J. (2017). Training confidence-calibrated classifiers for detecting out-of-
488 distribution samples. *arXiv preprint arXiv:1711.09325*.

489 Lee, K., Lee, K., Lee, H., and Shin, J. (2018a). A simple unified framework for detecting out-of-distribution
490 samples and adversarial attacks. *Advances in neural information processing systems*, 31.

491 Lee, K., Lee, K., Lee, H., and Shin, J. (2018b). A simple unified framework for detecting out-of-distribution
492 samples and adversarial attacks. In Bengio, S., Wallach, H., Larochelle, H., Grauman, K., Cesa-Bianchi, N.,
493 and Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 31. Curran Associates,
494 Inc.

495 Letouzey, F., Denis, F., and Gilleron, R. (2000). Learning from positive and unlabeled examples. In *International*
496 *Conference on Algorithmic Learning Theory*, pages 71–85. Springer.

497 Liang, S., Li, Y., and Srikant, R. (2018). Enhancing the reliability of out-of-distribution image detection in
498 neural networks. In *International Conference on Learning Representations*.

499 Liu, J., Lin, Z., Padhy, S., Tran, D., Bedrax Weiss, T., and Lakshminarayanan, B. (2020a). Simple and principled
500 uncertainty estimation with deterministic deep learning via distance awareness. In Larochelle, H., Ranzato,
501 M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing Systems*,
502 volume 33, pages 7498–7512. Curran Associates, Inc.

503 Liu, W., Wang, X., Owens, J., and Li, Y. (2020b). Energy-based out-of-distribution detection. *Advances in*
504 *neural information processing systems*, 33:21464–21475.

505 Liu, W., Wang, X., Owens, J., and Li, Y. (2020c). Energy-based out-of-distribution detection. In Larochelle,
506 H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing*
507 *Systems*, volume 33, pages 21464–21475. Curran Associates, Inc.

508 Maddox, W. J., Izmailov, P., Garipov, T., Vetrov, D. P., and Wilson, A. G. (2019). A simple baseline for bayesian
509 uncertainty in deep learning. In Wallach, H., Larochelle, H., Beygelzimer, A., d'Alché-Buc, F., Fox, E., and
510 Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

511 Malinin, A. and Gales, M. (2018). Predictive uncertainty estimation via prior networks. In Bengio, S., Wallach,
512 H., Larochelle, H., Grauman, K., Cesa-Bianchi, N., and Garnett, R., editors, *Advances in Neural Information*
513 *Processing Systems*, volume 31. Curran Associates, Inc.

514 Malinin, A. and Gales, M. (2019). Reverse kl-divergence training of prior networks: Improved uncertainty and
515 adversarial robustness. In Wallach, H., Larochelle, H., Beygelzimer, A., d'Alché-Buc, F., Fox, E., and Garnett,
516 R., editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

517 Meinke, A. and Hein, M. (2020). Towards neural networks that provably know when they don't know. In
518 *International Conference on Learning Representations*.

519 Ming, Y., Fan, Y., and Li, Y. (2022). POEM: Out-of-distribution detection with posterior sampling. In
520 Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S., editors, *Proceedings of the 39th*
521 *International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*,
522 pages 15650–15665. PMLR.

523 Ming, Y. and Li, Y. (2024). How does fine-tuning impact out-of-distribution detection for vision-language
524 models? *International Journal of Computer Vision*, 132(2):596–609.

525 Ming, Y., Sun, Y., Dia, O., and Li, Y. (2023). How to exploit hyperspherical embeddings for out-of-distribution
526 detection? In *The Eleventh International Conference on Learning Representations*.

527 Netzer, Y., Wang, T., Coates, A., Bissacco, A., Wu, B., Ng, A. Y., et al. (2011). Reading digits in natural images
528 with unsupervised feature learning. In *NIPS workshop on deep learning and unsupervised feature learning*,
529 volume 2011, page 4. Granada.

530 Nguyen, A., Yosinski, J., and Clune, J. (2015). Deep neural networks are easily fooled: High confidence
531 predictions for unrecognizable images. In *Proceedings of the IEEE conference on computer vision and pattern*
532 *recognition*, pages 427–436.

533 Niu, G., Du Plessis, M. C., Sakai, T., Ma, Y., and Sugiyama, M. (2016). Theoretical comparisons of positive-
534 unlabeled learning against positive-negative learning. *Advances in neural information processing systems*,
535 29.

536 Peng, B., Luo, Y., Zhang, Y., Li, Y., and Fang, Z. (2024). Conjnorm: Tractable density estimation for out-of-
537 distribution detection. In *The Twelfth International Conference on Learning Representations*.

538 Plessis, M. D., Niu, G., and Sugiyama, M. (2015). Convex formulation for learning from positive and unlabeled
539 data. In Bach, F. and Blei, D., editors, *Proceedings of the 32nd International Conference on Machine Learning*,
540 volume 37 of *Proceedings of Machine Learning Research*, pages 1386–1394, Lille, France. PMLR.

Ren, J., Fort, S., Liu, J., Roy, A. G., Padhy, S., and Lakshminarayanan, B. (2021). A simple fix to mahalanobis distance for improving near-ood detection. *arXiv preprint arXiv:2106.09022*.

Ren, J., Luo, J., Zhao, Y., Krishna, K., Saleh, M., Lakshminarayanan, B., and Liu, P. J. (2022). Out-of-distribution detection and selective generation for conditional language models. In *The Eleventh International Conference on Learning Representations*.

Saito, K., Kim, D., and Saenko, K. (2021). Openmatch: Open-set semi-supervised learning with open-set consistency regularization. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W., editors, *Advances in Neural Information Processing Systems*, volume 34, pages 25956–25967. Curran Associates, Inc.

Sehwag, V., Chiang, M., and Mittal, P. (2021). {SSD}: A unified framework for self-supervised outlier detection. In *International Conference on Learning Representations*.

Sun, Y., Guo, C., and Li, Y. (2021). React: Out-of-distribution detection with rectified activations. *Advances in Neural Information Processing Systems*, 34:144–157.

Sun, Y. and Li, Y. (2022). Dice: Leveraging sparsification for out-of-distribution detection. In *European Conference on Computer Vision*, pages 691–708. Springer.

Sun, Y., Ming, Y., Zhu, X., and Li, Y. (2022). Out-of-distribution detection with deep nearest neighbors. In *International Conference on Machine Learning*, pages 20827–20840. PMLR.

Tack, J., Mo, S., Jeong, J., and Shin, J. (2020a). Csi: Novelty detection via contrastive learning on distributionally shifted instances. *Advances in neural information processing systems*, 33:11839–11852.

Tack, J., Mo, S., Jeong, J., and Shin, J. (2020b). Csi: Novelty detection via contrastive learning on distributionally shifted instances. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing Systems*, volume 33, pages 11839–11852. Curran Associates, Inc.

Tao, L., Du, X., Zhu, J., and Li, Y. (2023). Non-parametric outlier synthesis. In *The Eleventh International Conference on Learning Representations*.

Uppaal, R., Hu, J., and Li, Y. (2023). Is fine-tuning needed? pre-trained language models are near perfect for out-of-domain detection. In Rogers, A., Boyd-Graber, J., and Okazaki, N., editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12813–12832, Toronto, Canada. Association for Computational Linguistics.

Van Amersfoort, J., Smith, L., Teh, Y. W., and Gal, Y. (2020). Uncertainty estimation using a single deep deterministic neural network. In *International conference on machine learning*, pages 9690–9700. PMLR.

Wang, H., Li, Z., Feng, L., and Zhang, W. (2022a). Vim: Out-of-distribution with virtual-logit matching. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4921–4930.

Wang, H., Liu, W., Bocchieri, A., and Li, Y. (2021). Can multi-label classification networks know what they don’t know? *Advances in Neural Information Processing Systems*, 34:29074–29087.

Wang, Q., Fang, Z., Zhang, Y., Liu, F., Li, Y., and Han, B. (2023a). Learning to augment distributions for out-of-distribution detection. In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine, S., editors, *Advances in Neural Information Processing Systems*, volume 36, pages 73274–73286. Curran Associates, Inc.

Wang, Q., Liu, F., Zhang, Y., Zhang, J., Gong, C., Liu, T., and Han, B. (2022b). Watermarking for out-of-distribution detection. *Advances in Neural Information Processing Systems*, 35:15545–15557.

Wang, Q., Ye, J., Liu, F., Dai, Q., Kalander, M., Liu, T., HAO, J., and Han, B. (2023b). Out-of-distribution detection with implicit outlier transformation. In *The Eleventh International Conference on Learning Representations*.

Wang, Y., Zou, J., Lin, J., Ling, Q., Pan, Y., Yao, T., and Mei, T. (2022c). Out-of-distribution detection via conditional kernel independence model. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural Information Processing Systems*, volume 35, pages 36411–36425. Curran Associates, Inc.

Wang, Z., Xiang, L., Huang, L., Mao, J., Xiao, L., and Yamasaki, T. (2024). Scomatch: Alleviating overtrusting in open-set semi-supervised learning. In *European Conference on Computer Vision*, pages 217–233. Springer.

Wei, H., Xie, R., Cheng, H., Feng, L., An, B., and Li, Y. (2022). Mitigating neural network overconfidence with logit normalization. In Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S., editors, *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 23631–23644. PMLR.

Wen, Y., Tran, D., and Ba, J. (2020). Batchensemble: an alternative approach to efficient ensemble and lifelong learning. In *International Conference on Learning Representations*.

Wilk, M. B. and Gnanadesikan, R. (1968). Probability plotting methods for the analysis of data. *Biometrika*, 55(1):1–17.

Wu, Q., Chen, Y., Yang, C., and Yan, J. (2023). Energy-based out-of-distribution detection for graph neural networks. In *The Eleventh International Conference on Learning Representations*.

Xu, D. and Denil, M. (2021). Positive-unlabeled reward learning. In *Conference on Robot Learning*, pages 205–219. PMLR.

Yang, J., Wang, H., Feng, L., Yan, X., Zheng, H., Zhang, W., and Liu, Z. (2021). Semantically coherent out-of-distribution detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 8301–8309.

Yang, J., Wang, P., Zou, D., Zhou, Z., Ding, K., Peng, W., Wang, H., Chen, G., Li, B., Sun, Y., et al. (2022). Openood: Benchmarking generalized out-of-distribution detection. *Advances in Neural Information Processing Systems*, 35:32598–32611.

Yang, J., Zhou, K., Li, Y., and Liu, Z. (2024). Generalized out-of-distribution detection: A survey. *International Journal of Computer Vision*, 132(12):5635–5662.

Yu, F., Seff, A., Zhang, Y., Song, S., Funkhouser, T., and Xiao, J. (2015). Lsun: Construction of a large-scale image dataset using deep learning with humans in the loop. *arXiv preprint arXiv:1506.03365*.

Zagoruyko, S. (2016). Wide residual networks. *arXiv preprint arXiv:1605.07146*.

Zhang, J., Yang, J., Wang, P., Wang, H., Lin, Y., Zhang, H., Sun, Y., Du, X., Zhou, K., Zhang, W., et al. (2023). Openood v1. 5: Enhanced benchmark for out-of-distribution detection. *arXiv preprint arXiv:2306.09301*.

Zhang, Y., Lu, J., Peng, B., Fang, Z., and ming Cheung, Y. (2024). Learning to shape in-distribution feature space for out-of-distribution detection. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*.

Zhao, Y., Xu, Q., Jiang, Y., Wen, P., and Huang, Q. (2022). Dist-pu: Positive-unlabeled learning from a label distribution perspective. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14461–14470.

Zheng, H., Wang, Q., Fang, Z., Xia, X., Liu, F., Liu, T., and Han, B. (2023). Out-of-distribution detection learning with unreliable out-of-distribution sources. *Advances in Neural Information Processing Systems*, 36:72110–72123.

Zhou, B., Lapedriza, A., Khosla, A., Oliva, A., and Torralba, A. (2017). Places: A 10 million image database for scene recognition. *IEEE transactions on pattern analysis and machine intelligence*, 40(6):1452–1464.

Zhou, Z., Guo, L.-Z., Cheng, Z., Li, Y.-F., and Pu, S. (2021). Step: Out-of-distribution detection in the presence of limited in-distribution labeled data. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W., editors, *Advances in Neural Information Processing Systems*, volume 34, pages 29168–29180. Curran Associates, Inc.

Zhu, J., Geng, Y., Yao, J., Liu, T., Niu, G., Sugiyama, M., and Han, B. (2023). Diversified outlier exposure for out-of-distribution detection via informative extrapolation. *Advances in Neural Information Processing Systems*, 36:22702–22734.

Zhu, Y., Chen, Y., Xie, C., Li, X., Zhang, R., Xue, H., Tian, X., zheng, b., and Chen, Y. (2022). Boosting out-of-distribution detection with typical features. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural Information Processing Systems*, volume 35, pages 20758–20769. Curran Associates, Inc.

A Additional Studies

A.1 Element-wise Median vs Geometric Median

Another pertinent question to ask is why we chose the element-wise median over the geometric median (Acharya et al., 2024) for outlier detection in Medix. To answer this question, we conducted a preliminary experiment using CIFAR-10 as the InD dataset and SVHN as the OOD dataset. Specifically, $\mathcal{S}_{\text{wild}}$ consists of 5k samples drawn from CIFAR-10, ensuring that these samples are disjoint from the training set used to train the model $\phi_{\mathcal{S}_{\text{in}}}$, which we leverage to compute $\bar{\nabla}_{\text{in}}$. We incrementally add SVHN OOD samples to $\mathcal{S}_{\text{wild}}$ and track the behavior of the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median of the gradients of the wild dataset as well as the proportion of OOD samples removed. The results, shown in the Figure 3, demonstrate that for the same number of OOD samples in the wild dataset, the element-wise median identified a significantly higher proportion of OOD samples as outliers compared to the geometric median. This indicates that the element-wise median is more sensitive to distributional shifts, making it a more effective and reliable choice for filtering in our method.

Table 3: Effect of hyperparameters ϵ and k on OOD detection.

Method	FPR95↓	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

A.2 Hyperparameter Selection and Sensitivity Analysis

We conducted an ablation study to assess the sensitivity of Medix to the hyperparameters ϵ and k . For this experiment, we employed CIFAR-100 as the InD dataset and SVHN as the OOD dataset. As shown in Table 3, Medix exhibits strong robustness to variations in the values of ϵ and k . In particular, Medix achieves a remarkably low FPR95 of 0.16 when using $\epsilon = 0.005$ and $k = 20000$. Even when the values of ϵ and k are varied—e.g., reducing ϵ to 0.0005 or halving k to 10000—the performance remains competitive (FPR95 of 0.20 and 0.68, respectively), demonstrating a graceful degradation rather than a sharp drop. The results indicate that while optimal hyperparameter selection is important, Medix maintains its effectiveness across a variety of hyperparameter choices, surpassing the baselines. This insensitivity to exact hyperparameter tuning makes Medix a reliable choice in real-world deployments, where exhaustive tuning may not be feasible.

A.3 Comprehensive comparison with recent competitive methods

We have extended our comparisons to include competitive baselines that employ diverse contrasting learning objectives, such as SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER Ming et al. (2023). Although these methods use contrasting learning objectives, which we do not, we included them for a comprehensive comparison. Additionally, we included more recent methods, such as CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al., 2022a), and VOS (Du et al., 2022c), to provide a more thorough evaluation. For this experiment, we use CIFAR-100 as InD

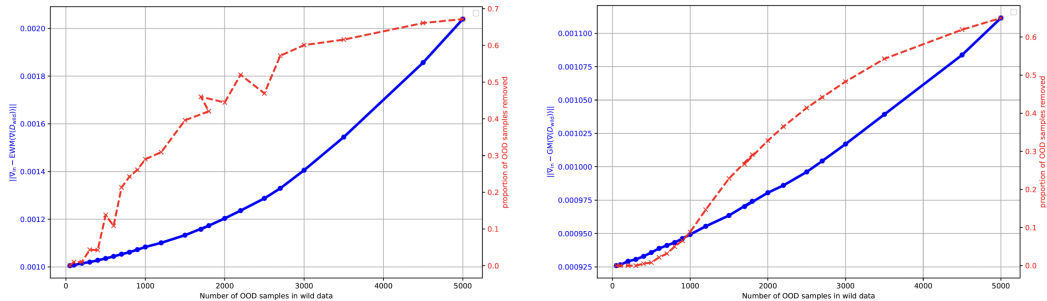


Figure 3: Comparison of element-wise median (EWM) and geometric median (GM).

Table 4: OOD detection performance comparison of Medix and recent competitive baselines on CIFAR-100 as InD data. Performance averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

dataset and train the ResNet-34 model¹ following the setup in Ming et al. (2023); Zhang et al. (2024). The model is trained using stochastic gradient descent with momentum 0.9, and weight decay 10^{-4} for 500 epochs. The initial learning rate is 0.5 with cosine scheduling and the batch size is 512. We use the checkpoints provided by Ming et al. (2023)². The results, as shown in Table 4, demonstrate that Medix outperforms even these competitive baselines by a significant margin, including those trained with contrastive learning objectives, achieving an average FPR95 of 8.68% and an average AUROC of 97.76%. Furthermore, Medix provides a provably low error rate, offering theoretical guarantees that most alternative approaches do not.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We next investigate whether Medix can handle large-scale wild OOD data under the unseen OOD setting. This setting is more challenging for two main reasons: (1) it leverages a large-scale wild OOD dataset ($\mathbb{P}_{\text{out}}$), increasing both data complexity and computational demand, and (2) the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is intentionally chosen to be distributionally different from the wild OOD data used during training, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$, which better reflects realistic and challenging deployment scenarios. We used CIFAR-100 as the labeled InD data, 300K Random Images dataset Hendrycks et al. (2019) as the unlabeled wild OOD data and tested on the SVHN dataset as the test OOD data. This setup introduces a greater level of complexity because the large-scale wild OOD data (300K Random Images) is significantly different from the OOD test data (SVHN). To evaluate the performance of our approach, we compare it against baselines that also leverage wild data in training. The results in the Table 5 highlight the superior performance of our method, Medix, compared to the baselines, with a significantly lower FPR95 (41.29 ± 1.2) and higher AUROC (87.25 ± 0.6), showing its effectiveness in distinguishing between InD and OOD data.

Table 5: Comparison of OOD detection performance on large-scale unseen OOD data.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

Another important question to ask is whether the quality of pseudo-labels $\hat{y}_{\tilde{x}_i}$, particularly in terms of softmax confidence, affects the performance of OOD filtering, and whether a simple pre-filtering step that removes low-confidence pseudo-labels could improve the robustness of the model. To answer

¹Since ResNet-34 is the standard model across these works, we adopt the same model to ensure consistency and facilitate a fair comparison.

²<https://github.com>

this question, we conducted an experiment to evaluate how filtering low-confidence pseudo-labels affects OOD filtering. For this experiment, we used CIFAR-10 as the InD dataset and LSUN-Resize as the OOD dataset. Specifically, we computed the softmax probabilities for each pseudo-labeled sample and discarded those with low-confidence predictions, setting a threshold of 0.6. After filtering, we found that 15.98% of the samples were removed from the training set.

The results showed that there was virtually no difference in performance between the method with filtering and the method without filtering. Both methods yielded a FPR95 of 0.01%, but with filtering, AUROC increased slightly from 99.98% to 99.99%. Thus, removing low-confidence pseudo-labels doesn't significantly impact the robustness of the model or its ability to detect OOD samples, showing that our method is resilient to noisy or low-confidence labels, and further filtering steps are unlikely to yield meaningful improvements.

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We now turn to the question of whether Medix's filtering phase is computationally feasible and memory-efficient on large datasets. In response to this question, we conducted profiling experiments on an NVIDIA A100-SXM4-80GB GPU using approximately 15,000 unlabeled samples for each InD–OOD pair. For this experiment, we used CIFAR-10 and CIFAR-100 as the InD datasets and LSUN-Resize as the OOD dataset. As seen from the results in Table 6, the GPU memory usage remains modest, and the filtering phase completes within a tractable timeframe, even when handling thousands of unlabeled samples. This demonstrates that the Medix filtering process is computationally feasible and does not impose excessive memory overhead, even on large datasets.

Table 6: Profiling results for Medix filtering on NVIDIA A100 80GB GPU.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Our work differs from recent semi-supervised open-set recognition methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) in several key ways. For example, in OpenML Saito et al. (2021), the main problem is to handle outliers in semi-supervised learning (SSL) when training a standard classifier, whereas in our setting, the main challenge is to detect OOD samples from unlabeled wild data and train a dedicated OOD detector classifier. In other words, while SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) aim to improve classification despite outliers, our approach enhances OOD detection in an open-world setting where labeled OOD data is unavailable.

Secondly, these SSL methods aim to train a classifier that is robust to OOD samples in SSL, treating outliers as noise, while in our setting, the goal is to explicitly detect OOD samples and train an OOD detector. In other words, SSL methods focus on suppressing OOD samples during training to improve SSL performance, whereas Medix actively detects and leverages these OOD samples to build a more effective and reliable OOD detection system.

Lastly, the two approaches differ in their techniques: OpenMatch is based on consistency regularization, where the main idea is to enforce consistency across different stochastic transformations of the same input. This is mathematically modeled through soft constraints on the classifier's outputs, encouraging smooth decision boundaries. On the other hand, Medix relies on median-based filtering for outlier detection. The method uses statistical robustness properties of the median, leveraging its insensitivity to extreme values to identify potential OOD samples. The theoretical analysis derives error bounds based on the contamination effect (proportion of OOD samples) and the concentration effect (sub-Gaussian behavior of InD gradients), ensuring that OOD detection error remains low.

749 B Broader Impacts and Limitations

750 As machine learning continues to advance, tackling the challenges of OOD detection has become
751 essential for ensuring robust and reliable model performance in real-world applications. We introduce
752 a median-centric framework for OOD detection that enhances OOD handling and model safety.
753 The impact of our research goes beyond theoretical advancements, with practical applications in
754 healthcare, autonomous systems, and finance. By improving OOD detection, we address a key
755 challenge in model deployment, fostering greater trust and adoption of machine learning technologies.
756 Future work could explore integrating it with generative models for outlier synthesis or adapting it to
757 dynamic environments where OOD distributions evolve over time.

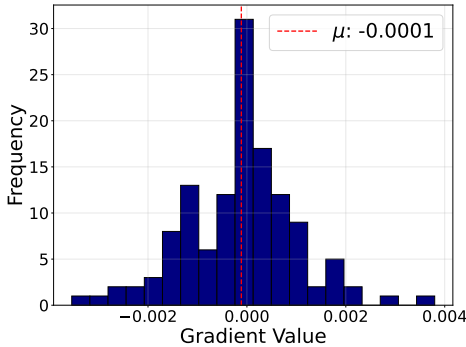
758 A potential limitation of Medix lies in its reliance on a mixture of unlabeled InD and OOD data,
759 which, in practice, may be noisy, corrupted, or inconsistently sampled. However, this assumption is
760 not unique to our method—it reflects the inherent uncertainty of real-world deployment environments
761 and is a common starting point in robust learning theory. To make our assumptions explicit and
762 verifiable, we rigorously formalize them in the main theorems and provide a precise geometric
763 condition in Remark 4.3 to justify the necessary separation required for reliable filtering. Moreover,
764 we extend our guarantees to looser settings in Theorem C.3, removing the need for sub-Gaussian tails
765 and demonstrating the resilience of our framework under weaker distributional assumptions. These
766 efforts underscore that while Medix does make simplifying assumptions, they are both interpretable
767 and relaxable, forming a principled foundation for future improvements in OOD detection under
768 realistic noise conditions.

769 **C Proof of Theorem**

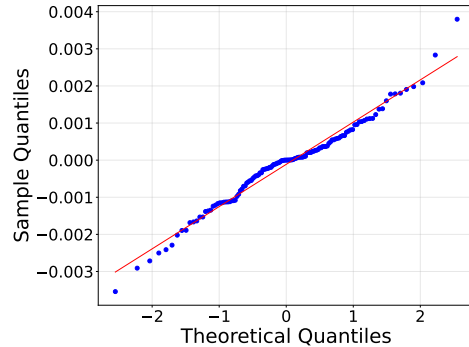
770 To aid clarity and consistency throughout the theoretical and empirical sections, we summarize the
 771 key notation used in this paper in Table 7. This includes definitions for gradient-related quantities,
 772 sample sizes, and filtering terms, as well as parameters that appear in our concentration bounds.
 773 Unless stated otherwise, we use bold symbols for vectors, $\mathcal{X}$ to denote the input space, and assume all
 774 gradients are computed with respect to a fixed pretrained model f_ϕ .

Table 7: Notation summary. We use bold symbols for vectors and $\mathcal{X}$ to denote the input space.

Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors



(a) InD gradient histogram.



(b) InD gradient Q-Q plot.

Figure 4: Illustration of InD sample gradients exhibiting sub-Gaussian behavior in each coordinate. (left) Histogram of gradient values (CIFAR-100 InD data) showing concentration around the mean with light tails, consistent with sub-Gaussianity. (right) Q-Q plot comparing empirical quantiles of InD gradients against a theoretical Gaussian distribution, confirming alignment with sub-Gaussianity.

775 C.1 Bound on the Inlier Misclassification Rate

776 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 777 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
 778 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 779 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

780 *Proof.* We structure our argument into two key steps. In **Step 1**, we establish a high-probability
 781 concentration bound for the gradients of InD samples around their mean, denoted as $\bar{\nabla}_{\text{in}}$. We define
 782 an InD sample whose gradient lies within this high-probability region as a *good inlier*. In **Step**
 783 **2**, we leverage a “swapping” argument on the EWM objective to demonstrate that removing more
 784 than a $(2\eta + \frac{\pi}{2}(1-\pi))$ fraction of *good inliers*—where η represents the fraction of InD samples
 785 with gradients outside the high-probability region (i.e., non-good InD samples)—would contradict
 786 optimality. Combining these two steps, we derive the stated upper bound on the inlier misclassification
 787 rate, ERR_{in} .

788 **Step 1: Concentration of InD Gradients.** Let $\nabla \ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ)
 789 of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with
 790 variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

791 Define

$$\mathbf{v} := \nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

792 It is straightforward to show that

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

793 so if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

794 Hence, combining (6) across $j = 1, \dots, d$ via a union bound yields

$$\mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

795 Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point
 796 $\mathbf{x}$ to be *bad* if

$$\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

797 Our goal is to show that, with high probability, only a small fraction of these m_{in} inliers can be bad.

798 From our sub-Gaussian assumption and applying the union bound across the d coordinates, we define
 799 a constant $p \in [0, 1]$ as follows:

$$p := \mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

800 Here, $\mathbf{x}$ is a generic InD sample, and we assume that the samples are i.i.d. across the m_{in} inliers.

801 Let X be the random variable denoting the number of bad inliers among the m_{in} samples. For each
 802 inlier $\mathbf{x}_i$, define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

803 Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

804 By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

805 Observing that X is a sum of m_{in} Bernoulli(p) random variables, each bounded between 0 and 1, we
806 can apply Hoeffding's inequality to obtain, for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

807 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log(\frac{1}{\delta})}{2}}, \quad (17)$$

808 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

809 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

810 Dividing both sides by m_{in} yields an upper bound on the fraction of bad inliers:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

811 Let us denote this fraction by η . By setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$ Thus, with probability at least
812 $1 - \delta$, no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients lying
813 outside the radius $\epsilon\sqrt{d}$ around $\bar{\nabla}_{\text{in}}$. These are precisely the *bad inliers*, while the remainder are
814 called *good inliers*.

815 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 816 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

817 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 818 thereby contradicting the optimality of $\mathcal{S}^*$.

819 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 820 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

821 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

822 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

823 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (26)$$

824 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 825 available for swapping in.

826 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 827 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

828 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

829 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

830 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

831 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

832 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 833 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

834 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

835 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

836 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

837 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

838 so the points lying inside B_ε form a strict majority in $\widehat{\mathcal{S}}$.

839 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
840 gradients in $\mathcal{S}^*$ and $\widehat{X}^{(j)} = \{\widehat{x}_1^{(j)}, \dots, \widehat{x}_m^{(j)}\}$ the same in $\widehat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

841 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
842 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
843 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\widehat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

844 which implies

$$|\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

845 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
846 increase it, so the median distance from μ_j does not increase and possibly decreases.

847 Thus, component-wise:

$$|\text{EWM}_j(G_{\widehat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

848 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

849 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
850 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

851 □

852 C.2 Bound on the Outlier Retention Rate

853 We now establish a non-asymptotic upper bound on the fraction of OOD points that remain in
854 the subset returned by the MEDIX filter. This result mirrors the inlier misclassification bound
855 (Appendix C.1), but instead of relying on inlier concentration, it exploits a separation assumption
856 between the OOD and InD gradient means.

857 **Theorem C.2** (Outlier Retention Bound under Vector Separation). Assume that for every OOD point
858 $x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:

859 **(I) Sub-Gaussianity.** Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .

860 **(II) Vector Mean Separation.** The mean OOD gradient vector μ_{out} satisfies

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d}, \quad \text{for some } \Delta > 0.$$

861 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

862 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

863 *Proof. Step 1: Concentration of OOD Gradients.*

864 By assumption, each coordinate of $g(x)$ is sub-Gaussian, and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
865 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

866 has probability at most $p_{\text{out}}(\varepsilon)$.

867 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

868 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

869 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

870 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

871 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

872 which holds with probability at least $1 - \delta$.

873 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

874 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_\varepsilon^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_\varepsilon^* > \frac{1-\pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

Thus, more than half the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients lie outside the ball

$$B_\varepsilon := \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (46)$$

By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of entries in $G_{\mathcal{S}_{\text{in}}^*}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now consider a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}}\eta_{\text{out}}$ are ambiguous, and the total number of InD points is m_{in} .

Then at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, implying:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

This contradicts the optimality of $\mathcal{S}_{\text{in}}^*$, and so the assumption must be false. Therefore, our assumption must be false. With probability at least $1 - \delta$, we conclude that the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1-\pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla\ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla\ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

895 **Differences with the original bound.** The original bound, which assumes sub-Gaussianity, features
 896 a decay term

$$2 \exp\left(-\frac{\epsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

897 that decreases exponentially with dimension d , making the bound extremely tight for large d . In
 898 contrast, the new bound, relying solely on the assumption of finite fourth moments, exhibits a decay
 899 term

$$\frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}$$

900 that decays as $O(1/d)$, yielding a looser bound for larger d . The original bound requires sub-Gaussian
 901 tails, while the new one only assumes finite fourth moments, allowing for heavier tails. In particular,
 902 both bounds share the finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

903 that shrinks as m_{in} increases, independent of the tail assumptions.

904 *Proof.* The proof follows the structure of the original Theorem 4.1, consisting of two main steps: (1)
 905 establishing a concentration bound on the fraction of “bad” InD samples (those far from ∇_{in}) without
 906 relying on sub-Gaussianity, and (2) using a swapping argument to bound the inlier misclassification
 907 rate. The key difference is the replacement of the sub-Gaussian tail bound with a moment-based
 908 bound derived from the finite fourth moment assumption.

909 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

910 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 911 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon \sqrt{d}, \quad (52)$$

912 and “good” otherwise. Our goal is to bound the fraction of bad inliers among the m_{in} InD samples in
 913 S_{wild} .

914 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we analyze the ℓ_2 norm via the sum of squared coordinates. Assume that
 915 the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$, each with:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

916 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

917 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

918 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

919 Using Chebyshev's inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

920 We need $\mathbb{P}(\|v\|_2 > \epsilon\sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

921 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

922 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

923 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
924 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

925 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
926 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

927 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

928 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

929 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

930 This η represents the upper bound on the fraction of InD samples whose gradients deviate from $\bar{\nabla}_{\text{in}}$
 931 by more than $\epsilon\sqrt{d}$, relying only on finite fourth moments and coordinate independence.

932 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 933 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

934 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 935 thereby contradicting the optimality of $\mathcal{S}^*$.

936 Let m_ϵ be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 937 lies outside the closed ℓ_2 ball

$$B_\epsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \epsilon\sqrt{d} \right\}. \quad (69)$$

938 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

939 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

940 Define the swap size as

$$m'_\epsilon := \min \{m_\epsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (71)$$

941 This quantity counts how many good inliers (those inside B_ϵ) are currently excluded from $\mathcal{S}^*$ and
 942 available for swapping in.

943 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ϵ points from $\mathcal{S}^*$ that lie outside B_ϵ (OOD points or bad inliers)
 944 and inserting m'_ϵ good inliers from B_ϵ in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

945 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\epsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

946 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\epsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

947 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

948 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

949 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 950 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

951 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

952 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta) m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

953 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

954 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

955 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

956 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
 957 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

958 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
 959 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
 960 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

961 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

962 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
 963 increase it, so the median distance from μ_j does not increase and possibly decreases.

964 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

965 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

966 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
 967 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

968 □